

- 3 A 2.0 kg block on a track is released at A, 1.0 m above the ground as shown in Fig. 3.1. The track is frictionless except for the rough surface between B and C, which has a length of 2.0 m. The block travels down the track, hits the spring of force constant $k = 225 \text{ N m}^{-1}$ at D and compresses the spring by 0.20 m from its equilibrium position before coming to rest momentarily.

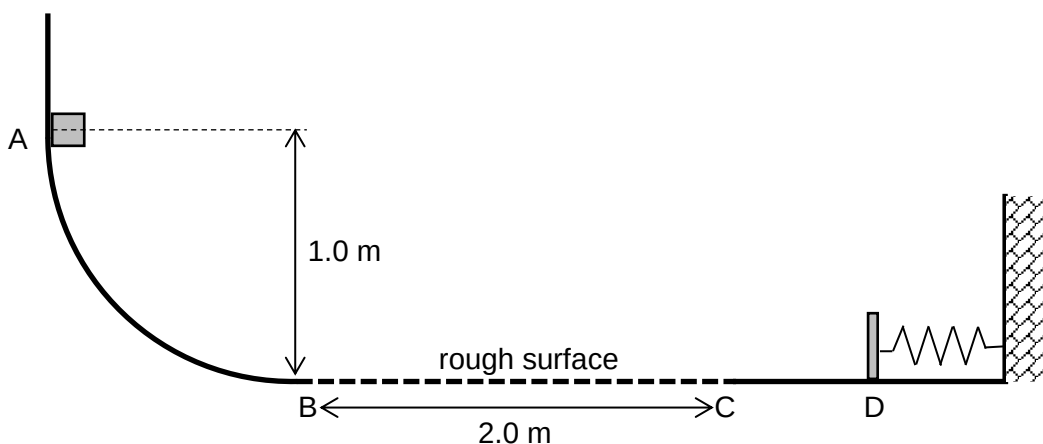


Fig. 3.1

- (a) (i) Determine the speed of the block at B.

By the Principle of conservation of energy:

Loss in gravitational E_p = Gain in E_k [M1]

$$mgh = \frac{1}{2}mv^2 - 0$$

$$v = \sqrt{2gh}$$

$$= \sqrt{2(9.81)(1.0)} = 4.43 \text{ m s}^{-1} \text{ [A1]}$$

speed at B = m s^{-1} [2]

- (ii) Calculate the maximum elastic potential energy stored in the spring.

$$\text{Energy stored in spring} = \frac{1}{2}kx^2$$

$$= \frac{1}{2}(225)(0.2)^2$$

$$= 4.5 \text{ J [A1]}$$

elastic potential energy = J [1]

- (iii) Using your answers to (a)(i) and (ii), determine the work done against friction when the block travels from B to C.

$$\text{Work done against friction} = E_{K(\text{initial})} - E_{P(\text{spring})}$$

$$= \frac{1}{2}(2.0)(4.43)^2 - 4.5 \text{ [M1]}$$

$$= 15.1 \text{ J [A1]}$$

work done against friction = J [2]

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- (b) The block subsequently rebounds and moves towards B after the spring un-compresses itself. Determine the distance along the track from C where the block finally stops.

$$\text{frictional force} = \frac{15.1}{2.0} = 7.56 \text{ N} \quad [\text{M1}]$$

If all the energy stored in the spring is used to do work against friction,

$$W = fd$$

$$4.5 = (7.56)d \quad [\text{M1}]$$

$$d = 0.596 \text{ m} \quad [\text{A1}]$$

distance from C = m [3]

[Total: 8]

4. A laser produces a narrow beam of coherent light of wavelength 632 nm. The beam is incident